Career taxonomy source and behavior inventory

Document ID: `SOURCE-INVENTORY-2026-09-01`
Audience: academic reviewer, product reviewer, and the next maintainer
Purpose: identify the authoritative source inputs, the repository's derived artifacts, and the currently implemented subject-matching behavior before the content and SQL release gates are frozen.

This inventory separates three claims that must not be confused:

1. a source exists and can be located;
2. the source says a particular fact; and
3. Sifter applies a documented rule to use that fact in a mapping or display.

A source URL or a file name alone proves only the first claim.

1. Authoritative source inputs

Nigerian admission and curriculum sources

Source layer	Local source input	What it can support	Repository evidence
JAMB/IBASS	`C:\Users\USER\Downloads\jamb brochure\` (10 brochure PDFs)	Programme catalogue names, national subject combinations, O-level wording, and direct-entry wording as printed	`data/career-mapping/_audit/jamb_brochure_source_audit_2026-08-17.json`
CCMAS/NUC	`C:\Users\USER\Downloads\ccmas original documents\` (17 original PDFs)	Programme overviews, objectives, courses, learning outcomes, and course content	`data/career-mapping/ccmas/*.json`, `data/career-mapping/_ccmas_txt/*.txt`
Nigerian context	Repository Nigerian professional-body, regulator, university, government, and local-sector source records	Nigerian terminology, credentials, regulatory boundaries, local practice, and local demand only where directly evidenced	`data/career-mapping/domain_role_source_registry.json` and the Nigerian enrichment catalogs

The JAMB brochure audit contains 416 checked catalogue rows: 400 exact matches, 16 declared aliases, and no unmatched catalogue row. It also records source file and page arrays. For example, `Medicine and Surgery (Human`

`Medicine)` is in `Medica_Pharmaceutical Health Science.pdf`, page 1. The printed-under-Law display variant of `Criminology and Security Studies` is linked to the same name in `Arts _ Humanities.pdf`, page 126, `Law.pdf`, page 13, and `Social Management Science.pdf`, pages 1, 36, 39, 41, 42, 44, 45, 49, and 50.

The extracted CCMAS text preserves printed page markers. For example, `data/career-mapping/_ccmas_txt/computing.txt` places the B.Sc. Computer Science overview, philosophy, and objectives at printed page 27 and the employability-skills section at printed page 28. The same source places `CSC 402: Ethics and Legal Issues in Computer Science` and its learning outcomes at printed page 50. The companion JSON keeps course-level locators such as `Computing::Computer Science::COS 201` and the extracted `skills_learned` text.

International classification and enrichment sources

Source	Supplied version/input	Proper role in the method	Key limitation
ISIC	Local ISIC Rev.5 workbook, `ISIC5_Exp_Notes_11Mar2024.xlsx`; SHA-256 `fd5986a1235259961787cce553c5515fb308b70b96580541246516ead5a7bdad`	Industry/activity context and completeness survey	Not an occupation, skill, or programme-suitability taxonomy
ISCO	Local `ISCO-08 EN Structure and definitions.xlsx`; SHA-256 `df10744491b4216ed3580e8263b6324502e774c16667a1d7ecdb...`	Occupational-function hierarchy and unit-group anchors	Does not define Sifter domain labels; raw absent codes remain explicit review findings
ESCO	Classification archive v1.2.1; SHA-256 `f9761b0b52be1828b38a3e5e0ae59a93a7b0ce358bc800f52f57a8b002adeb3e`	Preferred/alternative occupation labels, URIs, descriptions, and skills linked to ISCO	European/global vocabulary, not Nigerian demand proof
O*NET	Repository O*NET 31.0 and retained prior archive	Tasks, tools/software, work activities, preparation, and work context	US/global proxy; no automatic Nigerian counterpart
ILO	ISCO/ILO occupational material	International occupational definitions and employment context	Not direct Nigerian market evidence
WEF	Future of Jobs 2025 PDF; SHA-256 `8374fa3b3a55779d367ca6c295b6d6f1991caa1f01ece3c4cb40fb650115a223`	Trend and skills signals	Not a complete occupation list or local vacancy source

The domain definition library is the original classification input, not an official ISIC/ISCO replacement. Its SHA-256 is `de9637ce32ef5892932e47a11d57f87a3e76af5176c3ff5a1b4bce4f2485da03`. `data/career-mapping/_audit/domain_source_traceability_2026-08-31.json` records each domain's library row, raw anchors, label class, and source references. The current audit found D001-D166 in that library and D167-D168 as later repository additions.

2. Domain, subgroup, and role derivation

The derivation is layered rather than a claim that one source generated every name:

```
ISIC activity context
  -> source-library domain identity and boundary
  -> ISCO occupational-function anchors
  -> ESCO normalized occupations and skills
  -> O*NET/ILO/WEF task, preparation, and trend enrichment
  -> Nigerian sources for local terminology, regulation, credentials, and demand
  -> Sifter domain/subgroup/role display record
```

The source audit currently records:

Layer	Current audited scope	Important interpretation
Domain records	168 total; 160 visible	D001-D166 trace to the supplied library; D167-D168 are later curated additions
Domain label classes	7 exact source terms; 159 source-supported umbrellas; 2 curated additions	Practitioner-facing labels are not claimed to be verbatim official classification labels
Full subgroup provenance	555/555	The full taxonomy retains provenance; visible product scope is 542
Visible subgroup context	542/542	Each visible subgroup has at least one source-backed Nigerian context record
Merged roles	5,209	Source evidence may be exact, normalized, or source-role-only; these are not interchangeable claims
Visible role context	5,093/5,093	5,089 have actual local enrichment; four are explicitly global-only

When a subgroup cannot be defended, the product uses `domain -> role` rather than inventing a subgroup. A role without an exact Nigerian counterpart remains global-only or otherwise bounded; global evidence is not relabelled as local demand.

Layer-by-layer ownership

Product layer	Primary source facts	Enrichment sources	Current evidence artifact
Domain name and definition	Supplied domain-definition library; ISIC activity context; ISCO occupational anchors	ESCO, O*NET, ILO, WEF, specialist frameworks, Nigerian sector sources	`domain_meta.json`, `domain_source_traceability_2026-08-31.json`, naming provenance audit
Subgroup name and boundary	Domain library and ISCO sub-major/work-family structure	ESCO/O*NET occupation families, SFIA/NICE and specialist frameworks, Nigerian occupational standards	`domain_structure.json`, `computational_layer_manifest.json`, subgroup source traceability audit
Role identity and description	ISCO unit groups, ESCO occupations/URIs, O*NET occupation data and titles	WEF signals, specialist frameworks, Nigerian vacancy and regulator sources	`role_library_v3_cleaned.json`, `role_work_profiles.json`, `emerging_role_overlays.json`

Product layer	Primary source facts	Enrichment sources	Current evidence artifact
Skills and competencies	ESCO skill relations; O*NET Essential Skills, Transferable Skills, Work Activities, Tasks, Knowledge, and Abilities	SFIA, NICE, specialist frameworks, CCMAS learning outcomes where they describe programme preparation	`domain_skill_provenance.json`, `role_tools.json`, computational method evidence
Tools and software	O*NET Software Skills and named tools in checked role evidence	ESCO/tool relations and specialist technology frameworks; Nigerian sector sources for local context	`domain_tools.json`, `role_tools.json`, source-backed layer audit
Certifications and regulated preparation	Nigerian regulator/professional-body requirements and official credential routes	CFA, GARP, PMI, CIPS, SFIA, CPN, COREN, health councils, and other specialist bodies	certification gap audit, Nigerian context coverage report, learning-resource seed gate
Nigerian enrichment	Direct Nigerian regulator, professional-body, government, university, vacancy, and sector evidence	NBS labour context and dated vacancy/title signals	`nigerian_global_professional_overlays.json`, Nigerian context audit, role/subgroup gap overlays
Learning resources and source links	Source-backed reference pages, official frameworks, and learning/exam links	Resource catalog aggregation only; it does not create a new occupational fact	`learning_resource_population.manifest.json`, learning-resource population audit
Programme-to-domain relationship	Nigerian CCMAS courses, objectives, outcomes, and content; JAMB programme identity	Nigerian professional/local context for a bridge explanation; global sources do not repair weak curriculum fit	reviewed CCMAS/JAMB link artifact and active `jamb_domain_map.json`

The words “skill”, “competency”, “tool”, and “certification” are not treated as synonyms. A source may say that a role uses a tool or requires knowledge; that does not prove that a learner has demonstrated competence, and a certification resource does not prove that the certification is legally required in Nigeria. The display layer must preserve these distinctions.

3. Current JAMB/O-level matcher behavior

The executable behavior is in `backend/routes/student_pipeline_routes.py`, principally `programmes_by_subjects()` and `_match_structured_subject_options()`. The UI entry and state behavior are in `src/screens/careerIntelligence/AdmissionBrowseScreen.tsx`.

Three supported modes

Mode	Inputs	Rule applied	Result meaning
JAMB-only / `utme`	Exactly four normalized UTME subjects, including Use of English	Match only typed `utme` option records; required subjects and one-of groups must be satisfied	Subject-combination guidance for the JAMB gate, not an admission guarantee
O-level-only / `olevel`	Five to the configured early-guidance maximum normalized O-level subjects	Match only typed `olevel` option records; named requirements and the total credit minimum are enforced	O-level programme-fit guidance without using UTME rows
Combined / `combined`	Exactly four UTME subjects plus five to the O-level maximum	Independently match UTME and O-level by the same canonical JAMB programme ID, then intersect; retain O-level gaps/status	A combined view showing `eligible`, `utme_match_olevel_gap`, or `utme_match_olevel_unparsed` where applicable

The matcher fails closed unless the subject catalogue, structured option seed, and typed rule-record seed agree on programme ID, requirement ID, exam type, and source text. It does not turn a shared word in unparsed brochure prose into an exact match. University offerings are attached only after the canonical JAMB programme match; they add address, requirements, and notes rather than creating a different programme identity.

Worked matcher example

Suppose a typed UTME rule contains required `CHEM` and `BIO`, plus one of `MATH` or `PHY`, while its O-level rule requires `ENG`, `BIO`, `MATH`, and `CHEM`, plus one of `GEO`, `HSC`, `PHY`, or `STAT`.

```
JAMB-only selection:
  ENG, CHEM, BIO, PHY
  -> the UTME rule matches because CHEM and BIO are present and PHY satisfies
     the one-of group.

O-level-only selection:
  ENG, BIO, MATH, CHEM, GEO plus another subject
  -> the O-level rule matches because all named credits and the one-of group
     are present and the minimum subject count is met.

Combined selection:
  -> compute the two matches separately;
  -> keep the programme only if its same canonical JAMB programme ID occurs in
     both result sets;
  -> report any O-level missing subjects instead of flattening UTME and O-level
     into one list.
```

This is why the UI can say that a programme fits the UTME subjects while still showing an O-level gap. It is also why a stream changes ordering of subject choices but does not silently hide valid catalogue subjects. The independent matrix audit and tests document the same behavior in `docs/manual-review/subject-selection-matrix-run-2026-08-18.md` and `backend/tests/test_student_pipeline_match.py`.

4. Review and release interpretation

The historical CCMAS/JAMB complement audit reported 547 sufficient, 153 insufficient, and 11 missing-anchor occurrences. That was a before-state diagnostic. The current reviewed artifact covers all 711 occurrences:

Disposition	Count	Product meaning
Certified Nigerian curriculum link	547	Publish the relationship with exact curriculum anchors
Resolved Nigerian bridge link	163	Publish as a guided bridge, not an automatic qualification

Disposition	Count	Product meaning
Removed unsupported candidate	1	Keep in audit; do not seed/display as active

Therefore, the 153 were not silently erased or treated as exact matches. They were reclassified through the reviewed disposition layer. The remaining bounded states are also explicit: 220 intentional partial definitions, 135 no source-candidate records, 26 unmatched WEF signals, two ambiguous WEF signals, and four visible global-only roles. “Reviewed” means a disposition exists; it does not mean every item has an exact Nigerian counterpart, tool, certification, regulatory requirement, or local demand signal.

The population gate currently reports integrity success and permits SQL regeneration after the refreshed 5,795-row audit and review ledger were reconciled. The regenerated SQL remains a delivery artifact: it must be audited and applied only in a controlled Supabase run.

5. Complete taxonomy source register

The following is the complete source catalog currently routed by `domain_role_source_registry.json`. It is included here so an academic reader does not need repository access to know which source families were used. A catalog entry is a routing declaration; the evidence audits separately record whether the source was actually checked in an extract or source document.

Global, international, and specialist sources

Source ID	Source	Public location
`global_esco`	ESCO occupations and skills	ESCO
`global_onet`	O*NET 31.0 Database	O*NET
`global_isco`	ILO ISCO-08	ISCO
`global_wef`	World Economic Forum Future of Jobs 2025	WEF
`global_sfia`	SFIA 9	SFIA
`global_nice`	NIST NICE Workforce Framework	NICE
`global_supply_chain`	ASCM supply-chain body of knowledge	ASCM
`global_cips`	CIPS Global Standard for Procurement and Supply	CIPS
`global_project_management`	Project Management Institute standards	PMI
`global_health`	WHO health workforce resources	WHO
`global_fao`	FAO agriculture and food-systems resources	FAO
`global_woah`	WOAH animal-health standards	WOAH
`global_icao_iata`	ICAO and IATA aviation standards	ICAO / IATA
`global_imo_imarest`	IMO and IMarEST maritime standards	IMO / IMarEST
`global_built_environment`	RICS built-environment and property standards	RICS
`global_energy`	SPE petroleum-engineering competency resources	SPE
`global_nuclear`	IAEA nuclear and radiation resources	IAEA
`global_unesco`	UNESCO education, science, and culture resources	UNESCO
`global_creative`	ScreenSkills creative-industry role profiles	ScreenSkills
`global_game_dev`	International Game Developers Association resources	IGDA

Source ID	Source	Public location
tesda_psf_game_development	Philippine Skills Framework for Game Development	TESDA framework
global_xr	Khronos Group open standards	Khronos
qt_cfqt	European Competence Framework for Quantum Technologies v3.0	Quantum Flagship CFQT
nac_arts_technical_theatre	Singapore National Arts Council Technical Theatre and Production framework	Skills Framework
skills_england_live_event	Skills England Live Event Technician standard	Skills England
mqa_performing_arts	Malaysian Qualifications Agency Performing Arts standards	MQA
jff_xr	JFF XR and Immersive Learning Career Pathways Framework	JFF
arm_robotics_career	ARM RoboticsCareer competency pathways	RoboticsCareer
skills_england_robotics	Skills England Robotics Engineer occupational maps	Robotics Engineer / Advanced Robotics
british_esports	British Esports career pathways	British Esports
pearson_esports	Pearson BTEC Esports career pathways	Pearson BTEC
global_esports	Hitmarker esports and gaming jobs	Hitmarker
global_music	IFPI global music-industry resources	IFPI
global_wipo	WIPO intellectual-property resources	WIPO
global_cfa_finance	CFA Institute investment-profession resources	CFA Institute
global_garp_risk	Global Association of Risk Professionals resources	GARP
global_driver_training	DVSA National Standard for Driver and Rider Training	DVSA

Nigerian sources

Source ID	Source	Public location
ng_jobberman	Jobberman Nigeria jobs	Jobberman
ng_myjobmag	MyJobMag Nigeria feeds	MyJobMag
ng_nbs	National Bureau of Statistics Labour Force Survey	NBS microdata
ng_national_ai	Nigeria National Artificial Intelligence Strategy 2025	Local checked PDF in source registry
ng_nbte	NBTE National Occupational Standards and curricula	NBTE downloads
ng_cpn_syllabus_2021	CPN 2021 Examination Syllabus	CPN syllabus
ng_ncs	Nigeria Computer Society	NCS
ng_nsqf_nos	NBTE NSQF and National Occupational Standards	NSQF / PUCNOS
ng_cieeen	Council for Innovative Engineering and Emerging Engineering	CIEEEN
ng_health_ministry	Federal Ministry of Health and Social Welfare	Hospital services

Source ID	Source	Public location
ng_mls	Medical Laboratory Science Council of Nigeria	MLSCN documents
ng_clmis_corbon	CORBON construction labour-market information and NOS	CORBON CLMIS
ng_qsrbn	Quantity Surveyors Registration Board of Nigeria	QSRBN
ng_nnra	Nigerian Nuclear Regulatory Authority	NNRA
ng_trcn	Teachers Registration Council of Nigeria	TRCN
ng_nogicjqs	Nigerian Oil and Gas Industry Content Joint Qualification System	NOGIC JQS
ng_cmd	Centre for Management Development Nigeria	CMD
ng_nasrda	National Space Research and Development Agency	NASRDA
ng_nema	National Emergency Management Agency	NEMA
ng_fmacce	Federal Ministry of Arts, Culture and Creative Economy	FMaCCE
ng_ican_syllabus	Institute of Chartered Accountants of Nigeria 2026 syllabus	ICAN syllabus
ng_cibn	Chartered Institute of Bankers of Nigeria competency framework	CIBN
ng_nuj	Nigeria Union of Journalists professional ethics	NUJ ethics
ng_nla	Nigerian Library Association and LRCN context	NLA
ng_nicfost	Nigerian Council of Food Science and Technology	NICFOST
ng_nihotour	NIHOTOUR RIVAC registration portal	RIVAC
ng_fmafs	Federal Ministry of Agriculture and Food Security	FMAFS
ng_npa_psychology	Nigerian Psychological Association	NPA
ng_fashion_council	Nigerian Fashion Council and FADAN	FADAN
ng_sports_context	Nigeria Olympic Committee sports context	NOC
ng_public_health_context	NPHCDA and Nigerian public-health sources	NPHCDA
ng_nemet	Nigeria Meteorological Agency	NiMet
ng_nimet	Nigerian Meteorological Agency	NiMet
ng_nmdpra	Nigerian upstream and midstream petroleum regulators	NUPRC / NMDPRA
ng_legal_context	Nigerian legal, immigration, diplomacy and IP source set	NBA
ng_cipsmn	Chartered Institute of Purchasing and Supply Management of Nigeria	CIPSMN
ng_nas_research	Nigerian Academy of Science and national research sources	NAS
ng_nuc	National Universities Commission CCMAS and standards	NUC

Source ID	Source	Public location
`ng_cpn`	Computer Professionals Registration Council of Nigeria	CPN
`ng_telecom`	NCC and NITDA ICT ecosystem sources	NCC / NITDA
`ng_engineering`	COREN and Nigerian Society of Engineers	COREN / NSE
`ng_built_environment`	Nigerian built-environment regulators and bodies	ARCON / SURCON
`ng_health`	Nigerian health-profession regulators	MDCN / NMCN / PCN
`ng_finance`	Nigerian finance and accounting bodies/regulators	ICAN / SEC / CBN
`ng_business`	CIPM Nigeria and Nigerian Institute of Management	CIPM
`ng_supply_chain`	CILT Nigeria and Nigerian supply-chain bodies	CILT Nigeria
`ng_legal`	Nigerian Bar Association and legal regulators	NBA
`ng_education`	NUC, NCCE and Nigerian education standards	NCCE
`ng_agriculture`	Nigerian agriculture, food and veterinary authorities	NAFDAC / VCN
`ng_energy`	Nigerian upstream, midstream and downstream energy regulators	NUPRC / NMDPRA
`ng_aviation_maritime`	Nigerian aviation and maritime authorities	NCAA / NIMASA
`ng_creative`	Nigerian creative-industry and media sources	NUJ / NFVCB
`ng_gaming`	National Lottery Regulatory Commission and Nigerian gaming ecosystem	NLRC
`ng_environment`	NESREA and Nigerian environmental regulators	NESREA
`ng_surcon`	Surveyors Council of Nigeria	SURCON
`ng_mdcn`	Medical and Dental Council of Nigeria	MDCN
`ng_transport`	Federal Ministry of Transportation	Transportation Ministry
`ng_aviation`	Federal Ministry of Aviation and Aerospace Development	Aviation Ministry
`ng_choice_game_developer_2026`	Choice Talents Nigeria game-developer vacancy	Dated LinkedIn listing

Source files used for the Nigerian admission layer

The JAMB input set is the ten files in `C:\Users\USER\Downloads\jamb brochure`: `Administration.pdf`, `Agriculture.pdf`, `Arts _ Humanities.pdf`, `Education.pdf`, `Engineering _ Enviromental _ Technology.pdf`, `Law.pdf`, `Medica_Pharmaceutical Health Science.pdf`, `Sciences.pdf`, `Social Management Science.pdf`, and `Specific Entry Requirements.pdf`.

The CCMAS input set is the following seventeen original PDFs in `C:\Users\USER\Downloads\ccmas original documents`:

```
Administration-and-Management.pdf
Agriculture-2023.pdf
Allied-Health-Sciences-2023.pdf
Architecture-CCMAS-2023-FINAL.pdf
Arts-CCMAS-2023-FINAL.pdf
Basic-Medical-Sciences-CCMAS-FINAL-December-26-2022.pdf
Communication-and-Media-Studies-CCMAS-2023-FINAL.pdf
Computing-CCMAS-2023-FINAL.pdf
Education-CCMAS-2023-New.pdf
Engineering-Technology-CCMAS-2023-FINAL.pdf
Environmental-Sciences-CCMAS-2023-FINAL.pdf
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Law-ALL.pdf
Medicine-and-Dentistry-CCMAS-2023-FINAL.pdf
Pharmacy-and-Pharmaceutical-Sciences-CCMAS-2023-FINAL.pdf
Sciences-CCMAS-2023-FINAL.pdf
Social-Sciences-CCMAS-FINAL-2023-A.pdf
Veterinary-Medicine-CCMAS-2023-FINAL.pdf
```

Their exact file hashes are recorded in the source traceability packet and should be preserved with any future release packet. The extracted CCMAS text files are derived working extracts, not substitutes for the original PDFs.

6. External references

- JAMB FAQ — IBASS programme and UTME subject-combination guidance.
- IBASS education brochure
- IBASS sciences brochure
- IBASS arts brochure
- Official ISCO-08 page
- ISCO-08 Volume 1
- UN ISIC classification page
- ESCO use and classification resources

These links identify the public source. The repository audit artifacts identify the local file, version/hash, extracted fact, locator, and mapping decision.